

DIRECTIONS: THIS EXAMINATION CONTAINS A TOTAL OF **65** MULTIPLE CHOICE QUESTIONS. SELECT CAREFULLY THE BEST ANSWER AND BLACKEN ONLY THE LETTER OF YOUR CHOICE ON THE SEPARATE ANSWER SHEET PROVIDED

TIME ALLOWED: 3 HOURS

1. What is the equation of a circle whose end point of the diameter are (3, 1) and (-5,-3)?

A. $(x-1)^2 + (y+1)^2 = 20$

C. $(x-1)^2 + (y-1)^2 = 20$

B. $(x+1)^2 + (y+1)^2 = 20$

D. $(x+1)^2 + (y-1)^2 = 20$

2. If $x \neq \pm y$, then $\frac{\frac{x}{x+y} + \frac{y}{x-y}}{\frac{x}{x-y} - \frac{y}{x+y}} =$ _____

A. $\frac{x+y}{x-y}$

B. -1

C. 1

D. $\frac{x-y}{x+y}$

3. Which of the following is an anti-derivative of the function $f(x) = \frac{x+2}{x+1}$?

A. $1 + \ln|x+1| + c$

B. $1 \ln|x+2| + c$

C. $\ln\left|\frac{x+1}{x+2}\right| + C$

D. $x + \ln|x+1| + c$

4. Let p, q and r are three propositions whose truth values are F,T and F respectively, then which one of the following has a truth value false?

A. $(p \wedge q) \Rightarrow r \vee q$

B. $(r \vee p) \Rightarrow (p \Rightarrow r)$

C. $(p \Rightarrow r) \vee (p \wedge r)$

D. $(q \wedge p)$

5. Which one of the following is a divergent sequence?

A. $\left\{1 + \frac{(-1)^7}{7^n}\right\}$

B. $\left\{\frac{2n+1}{n-1}\right\}$

C. $\left\{\frac{2n}{n-1}\right\}$

D. $\left\{\left(\frac{7}{3}\right)^n\right\}$

6. The frequency distribution table of a grouped data x is given as shown below.

X	10 – 14	15 -19	20-24	25-29
Frequency	2	3	5	4

Which one of the following is true about this data?

A. The class mid- point of 3rd class is 22.5

C. the class mark of the 2nd class is 17.

B. The lower class limit of the 4th class is 24.5
the 3rd class is 24.

D. the upper class boundary of

7. If a point p does not lie on a line L in the coordinate plane, how far is the point p (-5,2) from the line $L: 3x+8y-5=0$

A. $\frac{5}{\sqrt{31}}$

B. $\frac{4}{\sqrt{73}}$

C. $\frac{12}{\sqrt{7}}$

D. $\frac{\sqrt{5}}{2}$

8. If the sequence recursively defined by $a_1=1$, $a_2=-2$ and $n \geq 3$. $a_n = a_{n-1} + (a_{n-2})^2 + n$, then the a_6 term is:
 A. 125 B. 175 C. 41 D. 21
9. If $\frac{5x+1}{x^2(x^2+1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{Cx+D}{x^2+1}$, where A,B,C and D are constants, then what are the values of A,B,C, and D respectively?
 A. $\frac{5}{4}, \frac{1}{4}, \frac{-5}{4}$ and $\frac{-1}{4}$ B. $\frac{-1}{4}, \frac{-5}{4}, \frac{1}{4}$ and $\frac{5}{4}$ C. $\frac{-5}{4}, \frac{-1}{4}, \frac{1}{4}$ and $\frac{5}{4}$ D. $\frac{1}{4}, \frac{5}{4}, \frac{-1}{4}$ and $\frac{-5}{4}$
10. Which of the following is NOT true about the function $f(x) = 3 - |x|$?
 A. The domain of $f = (-\infty, \infty)$. C. it is an even function
 B. The range of $f = [3, \infty)$ D. it's graph is symmetric with respect to the y-axis
11. Which of the following is null sequence?
 A. $\left\{ \frac{1-2n}{n+1} \right\}$ B. $\left\{ \cos\left(\frac{2}{n}\right) \right\}$ C. $\left\{ -(5)^{\frac{1}{n}} \right\}$ D. $\left\{ \frac{n+1}{1-n^2} \right\}$
12. Which one of the following compound propositions is a contradiction?
 A. $(p \vee q) \wedge (p \wedge q)$ B. $P \wedge (q \vee q)$ C. $(p \vee q) \wedge (p \vee q)$ D. $(p \wedge q) \wedge (p \vee q)$
13. The inverse of $f(x) = 1 - \ln(x+1)$ is _____
 A. $g(x) = e^{1-x} - 1$ B. $g(x) = e^{1-x} + 1$ C. $g(x) = 1 - e^{x-1}$ D. $g(x) = 1 - e^{1-x}$
14. the solution set of $\frac{2}{x} - \frac{x-2}{x^2-2x} = 1 - \frac{2x-2}{x-2}$ is _____
 A. $\{-1, 2\}$ B. $\{-2, 1\}$ C. $\{1\}$ D. $\{-2\}$
15. The value of $\int_0^1 (x+1)e^{x^2+2x} dx$ is _____
 A. $\frac{e^3}{2}$ B. $\frac{e^3-e}{2}$ C. $\frac{e^3-1}{2}$ D. $e^3 - 1$
16. Which one of the following function is a one to one correspondence?
 A. $f: [0, \infty) \rightarrow \mathbb{R}$ defined by $f(x) = |x|$ C. $f: \mathbb{R} \rightarrow [0, \infty)$ defined by $f(x) = 3^{x^2}$
 B. $f: \mathbb{R} \rightarrow [0, \infty)$ defined by $f(x) = x^2$ D. $f: (0, \infty) \rightarrow \mathbb{R}$ defined by $f(x) = \log_2 x$
17. If $x^2 + xy = 10$, then what is the value of $\frac{dy}{dx}$ when $x=2$?
 A. $-\frac{7}{2}$ B. $\frac{7}{2}$ C. $\frac{3}{2}$ D. $\frac{7}{2}$
18. In the following matrices A, B and C, if $AB=C$, then what is the value of x and y respectively?
 $A = \begin{pmatrix} -3 & 0 & 3 \\ 2 & x & -1 \\ 1 & 4 & -2 \end{pmatrix}$ $B = \begin{pmatrix} 5 & 2 & 6 \\ -1 & 1 & -3 \\ 0 & -3 & y \end{pmatrix}$ and $C = \begin{pmatrix} -15 & -15 & -6 \\ 11 & 6 & 11 \\ 1 & 12 & -14 \end{pmatrix}$
 A. 7 and -5 B. 3 and 5 C. -1 and -8 D. -1 and 4

19. What is the maximum value of the function $f(x) = x^4 - 2x^2$ on $[-2, 1]$?
 A. 8 B. 12 C. 24 D. 40
20. A satellite moves along a hyperbolic curve whose horizontal transverse axes is 24km and asymptote $y = \frac{5}{12}x + 2$. What is the eccentricity of the hyperbola?
 A. $\frac{5}{13}$ B. $\frac{12}{13}$ C. $\frac{13}{12}$ D. $\frac{15}{3}$
21. which one of the following is equal to $\lim_{x \rightarrow \infty} \left(\frac{x}{x+2} \right)^{-3x}$?
 A. e^6 B. e^{-3} C. $e^{\frac{3}{2}}$ D. e^{-6}
22. If $Z_1 = 5 - 3i$ and $Z_2 = 3 - 4i$, then $\frac{Z_1}{Z_2}$ is equal to _____
 A. $\frac{27}{25} - \frac{11}{25}i$ B. $\frac{27}{25} + \frac{29}{25}i$ C. $\frac{27}{25} + \frac{11}{25}i$ D. $\frac{3}{25} - \frac{11}{25}i$
23. The sum of all odd integer greater than 23 and less than 400 is given by: _____
 A. 212×189 B. 212×188 C. 424×189 D. 424×188
24. Which one of the following statements is simple statement?
 A. Neither integers nor whole numbers closed under division
 B. Both red fox and walia ibex are wild animals
 C. The number 2 is the only even prime number.
 D. It is not true that all odd number are not prime number
25. Which of the following is the negation of the statement “it is raining and weather is cold”?
 A. It is not raining and weather is cold C. it is not raining or weather is not cold
 B. It is raining or weather is not cold D. it is not raining and weather is not cold
26. What is the value of $\lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h}$?
 A. $\sin x$ B. $\cos x$ C. $-\sin x$ D. $\cot x$
27. If $f(x) = e^{2x} \sin x$, then the second derivative of $f(x)$ is equal to _____
 A. $-\sin x e^{2x}$ B. $-4e^{2x} \sin x$ C. $-e^{2x} \sin x - 2e^{2x} \cos x$ D. $3e^{2x} \sin x + 4e^{2x} \cos x$
28. If $a - b = 2c$, $a + b = 4c$ and $c = 3i + 4j$, then what are a and b respectively?
 A. $a = 9i - 12j$ and $b = 3i - j$ C. $a = 9i + 12j$ and $b = 3i + 4j$
 B. $a = 9i + 12j$ and $b = 3i + 4j$ D. $a = 9i - 12j$ and $b = -3i + 4j$
29. Let $f(x) = \sqrt{4 - x^2}$ then, which one of the following is not true about f ?
 A. f is continuous at each point in $[-2, 2]$. C. f is right continuous at $x = -2$
 A. f is left continuous at $x = 2$ D. f is left continuous at $x = -2$
30. Let $f(x) = x^3 - 12x + 1$ be defined on $[0, 3]$. what is the range of f ?
 A. $(-\infty, \infty)$ B. $(-\infty, 1]$ C. $[-15, 1]$ D. $[-15, 17]$
31. $\int \frac{x-2}{x^3 + 2x^2 - x - 2} dx$ equals to _____
 A. $\frac{3}{2} \ln|x+1| + \frac{4}{3} \ln|x+2| - \frac{1}{6} \ln|x-1| + C$ C. $\frac{3}{2} \ln|x+1| - \frac{4}{3} \ln|x+2| + \frac{1}{6} \ln|x-1| + C$

- B. $\frac{3}{2} \ln|x+1| - \frac{4}{3} \ln|x+2| - \frac{1}{6} \ln|x-1| + C$ D. $\frac{3}{2} \ln|x+1| + \frac{4}{3} \ln|x+2| + \frac{1}{6} \ln|x-1| + C$
32. Let $f(x) = 2-3x$ and $(f \circ g)(x) = 6x-1$. Then, $g(x)$ is equal to _____
A. $1-2x$ B. $2x-1$ C. $2x+1$ D. $2-x$
33. $\int e^x \sqrt{1+e^x} dx$ equals to _____
A. $\frac{2}{3} (1+e^x)^{\frac{3}{2}} + C$ B. $\frac{2}{3} (1+e^{2x})^{\frac{3}{2}} + C$ C. $\frac{2}{3} (1+e^x)^{\frac{3}{2}} + C$ D. $\frac{3}{2} (1+e^x)^{\frac{3}{2}} + C$
34. Let $L_1: 3x+2y=8$ $L_2: 4x+5y=1$ represent two intersecting line in a plane. If θ is the angle of inclination between L_1 and L_2 , what is $\tan \theta$?
A. $\frac{23}{2}$ B. $\frac{23}{5}$ C. $-\frac{23}{2}$ D. $-\frac{23}{5}$
35. $\int 3 \cos 3x dx$ is equal to _____
A. $3 \sin 3x + C$ B. $\sin 3x + C$ C. $-\sin 3x + C$ D. $\cos 3x + C$
36. For a given ellipse, what do you call the ratio of the distance between the two foci to the length of the major axis?
A. Directrix B. length of minor axis C. Eccentricity D. latus rectum
37. What is the derivative of the function $f(x) = \sqrt[5]{7^x}$?
A. $\frac{5}{\ln 7} \sqrt[5]{7^x}$ B. $\frac{\ln 5}{7} \sqrt[5]{7^x}$ C. $\frac{\ln 7}{5} \sqrt[5]{7^x}$ D. $\frac{\ln 7}{5} \sqrt[5]{7^x}$
38. What is the value of K so that $f(x) = \frac{x}{x^2+K}$ has a relative extremum at $x=2$?
A. 0 B. 2 C. 4 D. 8
39. What is the 3rd decile (D_3) of the data 2, 5, 3, 4, 11, 9, 8, 7, 12, 19?
A. 5.5 B. 4.5 C. 5 D. 4
40. Let $A = \begin{pmatrix} -4 & 2 & 0 \\ 1 & 3 & 6 \\ 5 & 7 & 9 \end{pmatrix}$. Which of the following is NOT true about this matrix?
A. Cofactor of $a_{23} = 38$ C. cofactor of $a_{13} = -8$
B. minor of $a_{13} = \text{cofactor of } a_{31}$ D. minor of $a_{23} = -38$
41. The second derivative of $f(x) = \ln(x^2+1)$ is _____
A. $\frac{1-2x^2}{x^2+1}$ B. $\frac{-2(x^2-1)}{x^2+1}$ C. $\frac{-2(x^2-1)}{(x^2+1)^2}$ D. $\frac{2-4x^2}{(x^2+1)^2}$
42. What is the value of K so that the line passing through the point $(K, -1)$ and $(2, -k)$ has slope $-\sqrt{2}$?
A. $3-\sqrt{2}$ B. $1-\sqrt{2}$ C. $3+\sqrt{2}$ D. $\sqrt{2}-3$
43. What is the absolute maximum value of $f(x) = 2x^2 - x^4$ on $[0, 2]$?
A. -3 B. 3 C. -4 D. 12

44. Among students who took a quiz, 15 students scored 6, 20 students scored 7, 10 students scored 8, 5 students scored 10. What is the average score of the students?
 A. 7.8 B. 7.5 C. 7.2 D. 7.0

1. Let $f(x) = \log_5 x$, $g(x) = x^2 + 2$ what is g of (x) ?

A. $(\log_5 x)^2 + 2$ B. $\log_5 (x^2 + 2)$ C. $(\log_5 x)^2 - 2$ D. None

45. $\frac{-1}{10} + \frac{1}{10}$ is equal to A. $\frac{-2}{10}$ B. $\frac{-1}{10}$ C. 0 D. $\frac{2}{10}$ If $x = 4$ & $y = -6$, then find $\frac{|x|}{x+1}$.

A. 5/4 B. 4/5 C. -6/5 D. 6/5

46. What is the greatest integer $x = 6$. A. 6 B. -5 C. -6 D. None

47. What is the solution set of $|x-3| = -1$? A. $x=2$ B. $x=-4$ C. $x=4$ D. $\{ \}$

48. Let $f(x) = \frac{x+1}{x}$, then $f(1) - f(2)$ is equal to A. 1 B. 2 C. 1/2 D. -1

49. Let $f(x) = 3x+5$ and $g(x) = 2x+1$, then $f \circ g(1)$ is equal to A. 13 B. 14 C. 12 D. 15

50. What is the subtraction of $\frac{3x+13}{x+4} - \frac{2x+7}{x+4}$ A. $\frac{x+6}{x+4}$ B. $\frac{x-6}{x+4}$ C. $\frac{x+4}{x+6}$ D. $\frac{x-4}{x-6}$

51. What is the distance from the origin to the line $5x-2y-7=0$?

A. $d = -7/\sqrt{29}$ B. $d = 7/\sqrt{29}$ C. $\sqrt{29}/7$ D. $-\sqrt{29}/7$

2. What is the tangent of the angle between the given lines? $L_1: y = -3x+2$, $L_2: y = -x$

A. $\tan \beta = 1/2$ B. $\tan \beta = -1/2$ C. $\tan \beta = -2$ D. $\tan \beta = 2$

52. Let $f(x) = \log_5 x$, $g(x) = x^2 + 2$ what is g of (x) ?

A. $(\log_5 x)^2 + 2$ B. $\log_5 (x^2 + 2)$ C. $(\log_5 x)^2 - 2$ D. None

53. What is the tangent of the angle between the given lines? $L_1: y = -3x+2$, $L_2: y = -x$

A. $\tan \beta = 1/2$ B. $\tan \beta = -1/2$ C. $\tan \beta = -2$ D. $\tan \beta = 2$

54. What is the determinate of the matrix $\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix}$

A. 37 B. 47 C. -47 D. -37

55. What is the modulus of the complex number $-6+8j$

A. 8 B. 2 C. 0 D. 4

56. The following sequences Converts of $\left\{ \frac{3n+4}{n} \right\}$

A. 3 B. 2 C. 0 D. 5

57. The equation of circle is given by $(k-3)^2 + (y+2)^2 = 9$ what are the center and radius of the circle respectively?

A. $(3, -2)$ and 9 B. $(3, 2)$ and 9 C. $(3, -2)$ and 3 D. $(3, 2)$ and 3

58. Consider the data given below

10, 12, 10, 15, 12, 16, 13, 11, 12, 15

The mean deviation about the mode of this data is equal to

A. 20 B. 1.8 C. 1.6 D. 1.5

59. Which Of The Following Is A Geometric Sequence?

A. 8,-4,2, -1, 1 ...

C. 1, 1.5, 2.5, 3.5,.....

B. 0,1,0,11,0.12,0.13....

D. 0.1, 0.01, 0.0011,0.000111...

60. What is the least upper bound of the sequence $\{2 + \frac{3}{n+1}\}_{n=1}$

A. 3 B. 2 C. 1 D. 0

61. Which of the following is the form of the partial fraction decomposition of

$$\frac{2x^3+2x^2+1}{(k+1)^3(x^2+1)}$$

A. $\frac{A}{K+1} + \frac{B}{(K+1)^2} + \frac{C}{x^2+1}$

c. $\frac{A}{X+1} + \frac{Bx+c}{(K+1)^2} + \frac{D}{x^2+1}$

B. $\frac{A}{X+1} + \frac{Bn+1}{(K+1)^2} + \frac{D}{x^2+1}$ D. $\frac{A}{X+1} + \frac{B}{(X+1)^2} + \frac{c}{x^2+1} + D$

62. When you divide the polynomial $f(x)=3x^3+2k^2-19x+6$ by $k-1$ the reminder that you will get is _____

A. -8 B. 8 C. 0 D. 24

63. Sin 25 is equal to _____

A. Sin 65° B. Cos65° C. $\frac{1}{\sec 65^\circ}$ D. $\frac{1}{\csc 65^\circ}$

64. Which of the following is product AB the two matrices

A. $\begin{bmatrix} -2 & 3 \\ 4 & -5 \end{bmatrix}$ and B = $\begin{bmatrix} -1 & 5 \\ 3 & 0 \end{bmatrix}$

A. $\begin{bmatrix} -11 & 10 \\ 11 & -5 \end{bmatrix}$

C. $\begin{bmatrix} -11 & 10 \\ 11 & 5 \end{bmatrix}$

B. $\begin{bmatrix} -11 & -10 \\ 11 & 5 \end{bmatrix}$

D. $\begin{bmatrix} -11 & 10 \\ 11 & 5 \end{bmatrix}$

65. Which of the following statements has a truth value T?

A. $(P=Q)=(PVQ)$ B. $(pvq)=(pnq)$ C. $(pnq)=>(pvq)$ D. $(PNQ)=(PVQ)[-11 \ 10]$

